

## PUERTO MONTT, Chile

WMO: 857990

Lat: 41.4350S

Lon: 73.0978W

Elev: 84

StdP: 100.32

Time Zone: -4.00 (CHL)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | -1.8       | -0.8 | -2.9                          | 3.0 | 0.4  | -1.7 | 3.3 | 0.4  | 12.5                  | 9.4  | 11.3 | 9.5  | 0.4                   | 300  | 0.585 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 10.4                   | 23.3            | 16.8 | 21.9 | 16.3 | 20.2 | 15.4 | 17.8                | 21.8 | 16.9 | 20.7 | 16.0 | 19.2 | 4.3                  | 150  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 16.1                            | 11.6 | 19.2 | 15.2 | 10.9 | 18.2 | 14.5 | 10.4 | 17.5 | 50.5          | 21.9 | 47.7 | 20.8 | 45.2 | 19.2 | 22.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 10.3          | 8.8  | 7.7 | -3.7                       | 28.1 | 0.9                | 2.1 | -4.4                                               | 29.7 | -4.9       | 30.9 | -5.5       | 32.1 | -6.1       | 33.6 |    |    |
| (5)               |      |     | -3.9                       | 19.7 | 1.0                | 1.1 | -4.6                                               | 20.5 | -5.2       | 21.1 | -5.8       | 21.7 | -6.5       | 22.6 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 10.4        | 14.3  | 14.2  | 12.7  | 10.4  | 9.1   | 7.2   | 6.6   | 7.3   | 8.2   | 9.8   | 11.3  | 13.2  |  |  |
|                                                                        | DBStd       | 3.55        | 2.17  | 2.37  | 2.26  | 2.49  | 2.68  | 2.68  | 2.59  | 2.35  | 2.19  | 1.99  | 2.10  | 2.17  |  |  |
|                                                                        | HDD10.0     | 467         | 1     | 1     | 3     | 24    | 49    | 90    | 108   | 88    | 62    | 29    | 10    | 2     |  |  |
|                                                                        | HDD18.3     | 2915        | 126   | 116   | 174   | 237   | 286   | 334   | 363   | 342   | 304   | 266   | 210   | 158   |  |  |
|                                                                        | CDD10.0     | 597         | 134   | 119   | 87    | 37    | 22    | 6     | 4     | 4     | 8     | 22    | 50    | 102   |  |  |
|                                                                        | CDD18.3     | 3           | 1     | 2     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 68          | 20    | 32    | 6     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 2     | 8     |  |  |
|                                                                        | CDH26.7     | 11          | 4     | 6     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     |  |  |
| (14) Wind                                                              | WSAvg       | 3.0         | 3.0   | 2.9   | 2.6   | 2.7   | 3.0   | 3.4   | 3.3   | 3.5   | 2.9   | 3.1   | 3.0   | 3.0   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 1698        | 82    | 82    | 95    | 139   | 204   | 214   | 202   | 198   | 133   | 126   | 107   | 103   |  |  |
|                                                                        | PrecMax     | 2287        | 246   | 252   | 192   | 269   | 442   | 362   | 418   | 378   | 247   | 288   | 236   | 311   |  |  |
|                                                                        | PrecMin     | 965         | 10    | 11    | 17    | 52    | 30    | 45    | 80    | 73    | 43    | 23    | 27    | 9     |  |  |
|                                                                        | PrecStd     | 260         | 50    | 52    | 38    | 60    | 94    | 70    | 77    | 68    | 47    | 56    | 45    | 67    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 25.7  | 27.2  | 23.8  | 20.9  | 17.6  | 14.2  | 14.0  | 15.2  | 17.9  | 20.0  | 21.8  | 24.2  |  |  |
|                                                                        |             | MCWB        | 18.4  | 17.2  | 17.1  | 15.9  | 14.0  | 12.2  | 11.8  | 11.3  | 12.0  | 13.6  | 14.8  | 16.9  |  |  |
|                                                                        | 2%          | DB          | 22.9  | 23.8  | 20.9  | 18.0  | 15.8  | 13.1  | 12.8  | 13.2  | 15.2  | 17.1  | 19.0  | 21.8  |  |  |
|                                                                        |             | MCWB        | 16.9  | 17.1  | 15.9  | 14.3  | 13.3  | 11.5  | 10.9  | 10.7  | 10.8  | 12.3  | 14.3  | 15.9  |  |  |
|                                                                        | 5%          | DB          | 21.1  | 21.8  | 19.1  | 16.2  | 14.3  | 12.2  | 11.8  | 12.2  | 13.8  | 15.4  | 17.2  | 20.0  |  |  |
|                                                                        |             | MCWB        | 16.0  | 16.7  | 15.0  | 13.7  | 12.5  | 10.8  | 10.3  | 10.2  | 10.4  | 11.6  | 13.0  | 15.1  |  |  |
|                                                                        | 10%         | DB          | 19.8  | 19.9  | 17.8  | 15.0  | 13.2  | 11.2  | 10.9  | 11.2  | 12.2  | 14.1  | 16.0  | 18.4  |  |  |
|                                                                        |             | MCWB        | 15.2  | 15.5  | 14.2  | 12.9  | 11.8  | 10.1  | 9.5   | 9.7   | 10.9  | 12.3  | 13.9  | 15.9  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 19.2  | 19.5  | 18.2  | 16.6  | 14.9  | 13.2  | 12.7  | 13.0  | 13.1  | 14.3  | 16.1  | 18.0  |  |  |
|                                                                        |             | MCDB        | 24.8  | 24.3  | 21.5  | 18.7  | 16.5  | 14.0  | 13.4  | 14.0  | 16.2  | 17.7  | 19.8  | 22.7  |  |  |
|                                                                        | 2%          | WB          | 17.6  | 18.0  | 16.7  | 15.2  | 13.9  | 12.1  | 11.5  | 11.7  | 11.8  | 13.1  | 14.8  | 16.6  |  |  |
|                                                                        |             | MCDB        | 21.7  | 22.1  | 19.6  | 17.2  | 15.2  | 12.9  | 12.3  | 12.8  | 14.1  | 15.9  | 17.9  | 20.7  |  |  |
|                                                                        | 5%          | WB          | 16.6  | 16.9  | 15.6  | 14.1  | 13.0  | 11.3  | 10.6  | 10.7  | 11.0  | 12.2  | 13.8  | 15.5  |  |  |
|                                                                        |             | MCDB        | 20.3  | 20.7  | 18.1  | 15.8  | 14.3  | 12.1  | 11.4  | 11.9  | 12.9  | 14.7  | 16.7  | 19.1  |  |  |
|                                                                        | 10%         | WB          | 15.6  | 15.8  | 14.8  | 13.2  | 12.2  | 10.5  | 9.8   | 9.9   | 10.3  | 11.4  | 12.8  | 14.6  |  |  |
|                                                                        |             | MCDB        | 18.8  | 19.1  | 17.1  | 14.7  | 13.3  | 11.3  | 10.5  | 10.9  | 11.8  | 13.6  | 15.3  | 17.7  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 10.4  | 10.5  | 9.9   | 8.7   | 7.4   | 6.5   | 6.9   | 7.3   | 8.6   | 8.8   | 9.1   | 9.8   |  |  |
|                                                                        |             | MCDBR       | 12.7  | 13.9  | 12.3  | 10.3  | 8.3   | 6.4   | 7.0   | 7.9   | 10.7  | 11.0  | 11.3  | 12.4  |  |  |
|                                                                        | 5% WB       | MCWBR       | 7.5   | 8.2   | 7.8   | 7.2   | 6.2   | 5.2   | 5.3   | 5.7   | 6.9   | 7.0   | 7.1   | 7.3   |  |  |
|                                                                        |             | MCWBR       | 11.5  | 12.4  | 10.6  | 9.1   | 7.1   | 5.5   | 5.4   | 6.1   | 8.7   | 9.5   | 10.3  | 11.3  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.332 | 0.343 | 0.328 | 0.320 | 0.319 | 0.318 | 0.315 | 0.325 | 0.329 | 0.327 | 0.325 | 0.329 |  |  |
|                                                                        |             | taud        | 2.537 | 2.500 | 2.549 | 2.554 | 2.539 | 2.529 | 2.537 | 2.508 | 2.478 | 2.506 | 2.533 | 2.536 |  |  |
|                                                                        | Ebn at Noon | Ebn at Noon | 996   | 958   | 929   | 867   | 794   | 756   | 789   | 846   | 911   | 964   | 997   | 1004  |  |  |
|                                                                        |             | Edh at Noon | 107   | 106   | 92    | 79    | 68    | 63    | 67    | 81    | 97    | 104   | 107   | 109   |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 6.81  | 5.84  | 4.26  | 2.71  | 1.64  | 1.19  | 1.45  | 2.09  | 3.33  | 4.46  | 5.67  | 6.51  |  |  |
|                                                                        | RadStd      | RadStd      | 0.53  | 0.45  | 0.38  | 0.25  | 0.22  | 0.17  | 0.13  | 0.23  | 0.39  | 0.47  | 0.53  | 0.84  |  |  |

## Historical Trends

|      |                       | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |      |
|------|-----------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|------|
|      |                       |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |      |
| (46) | Station Only          | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     | (46) |
|      |                       | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |      |
| (47) | Regional (1 neighbor) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     | (47) |

Nomenclature: See separate page